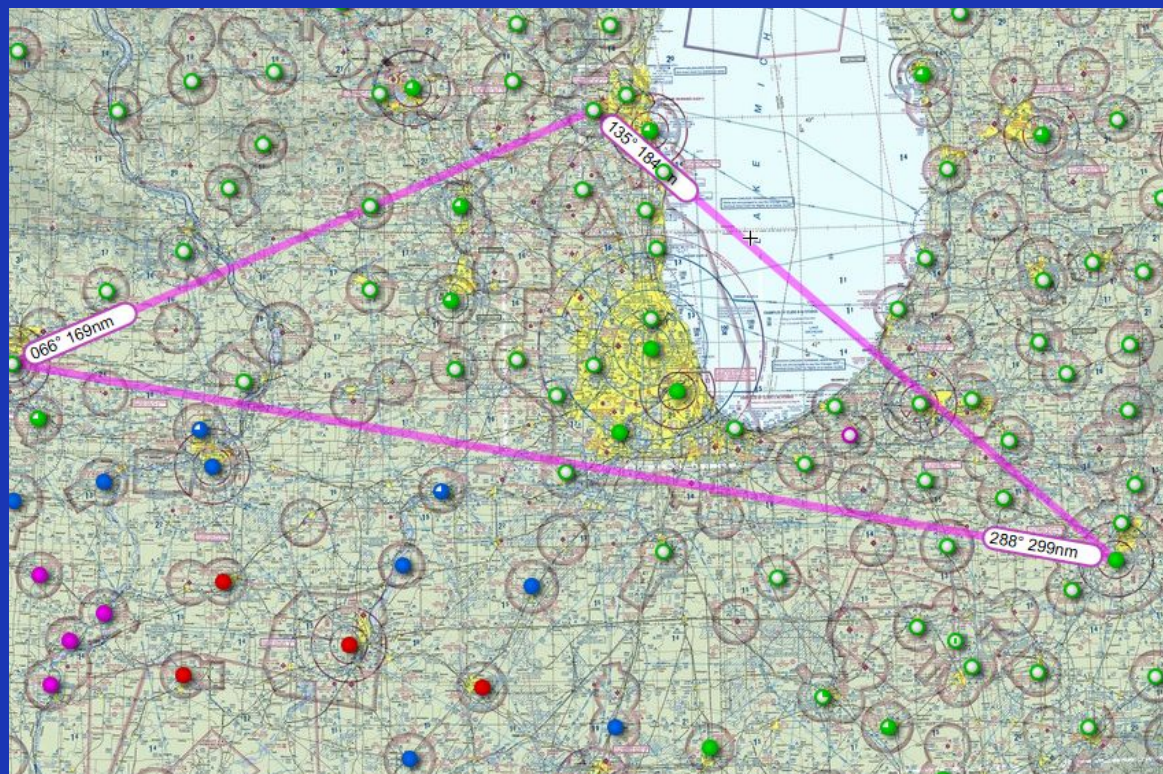


Activity 1.3.5

GPS Navigation

**By: Rhithvik
Selvakumar**

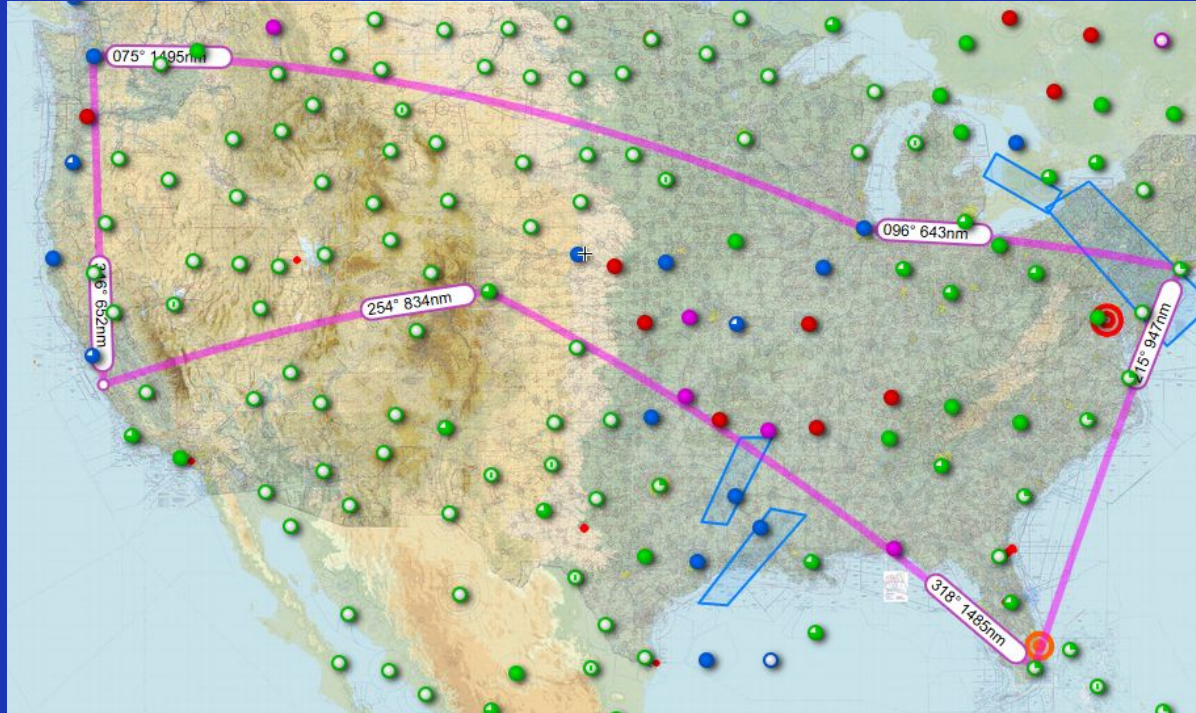


CESSNA 172 SPECIFICATIONS

Fuel capacity	Cruising air speed	Cruising Range
44 gal	120 kt	540 nm
Airport Name	Airport Identifier	VOR Frequency
Waukesha	KUES	116.4
Fort Wayne	KFWA	117.8
Eastern Iowa	KCID	114.1

Travel Segment	Distance (nm)	Total Distance Travelled (nm)	Airspeed (kt)	Travel time (hr)	TO vector	FROM vector
Waukesha-Fort Wayne	184	184	120	1.533333333	135	315
Fort Wayne-Eastern Iowa	299	483	120	2.491666667	288	108
Eastern Iowa-Waukesha	169	652	120	1.408333333	66	246

Flight Plan

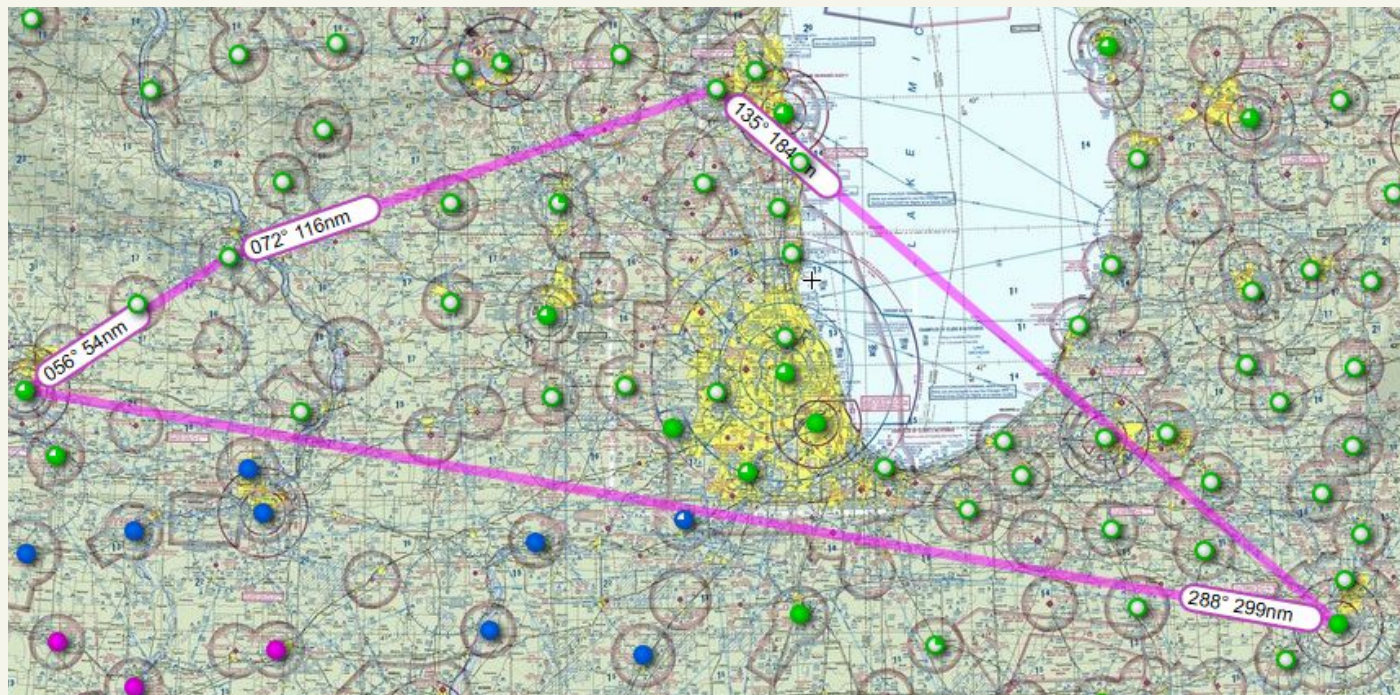


Airport Information

Airport Code	Airport Name	Latitude	Longitude	Elevation (MSL)
JFK	NYC JFK	N40°38.59'	W73°46.81'	13.0
MIA	Miami	N25°47.80'	W80°17.41'	9.3
DEN	Denver	N39°51.72'	W104°40.41'	5433.8
MRY	Monterey	N36°35.24'	W121°50.71'	256.6
SEA	Seattle	N47°27.48'	W122°18.65'	432.3
ORD	Chicago O'Hare	N41°58.66'	W87°54.31'	680.0

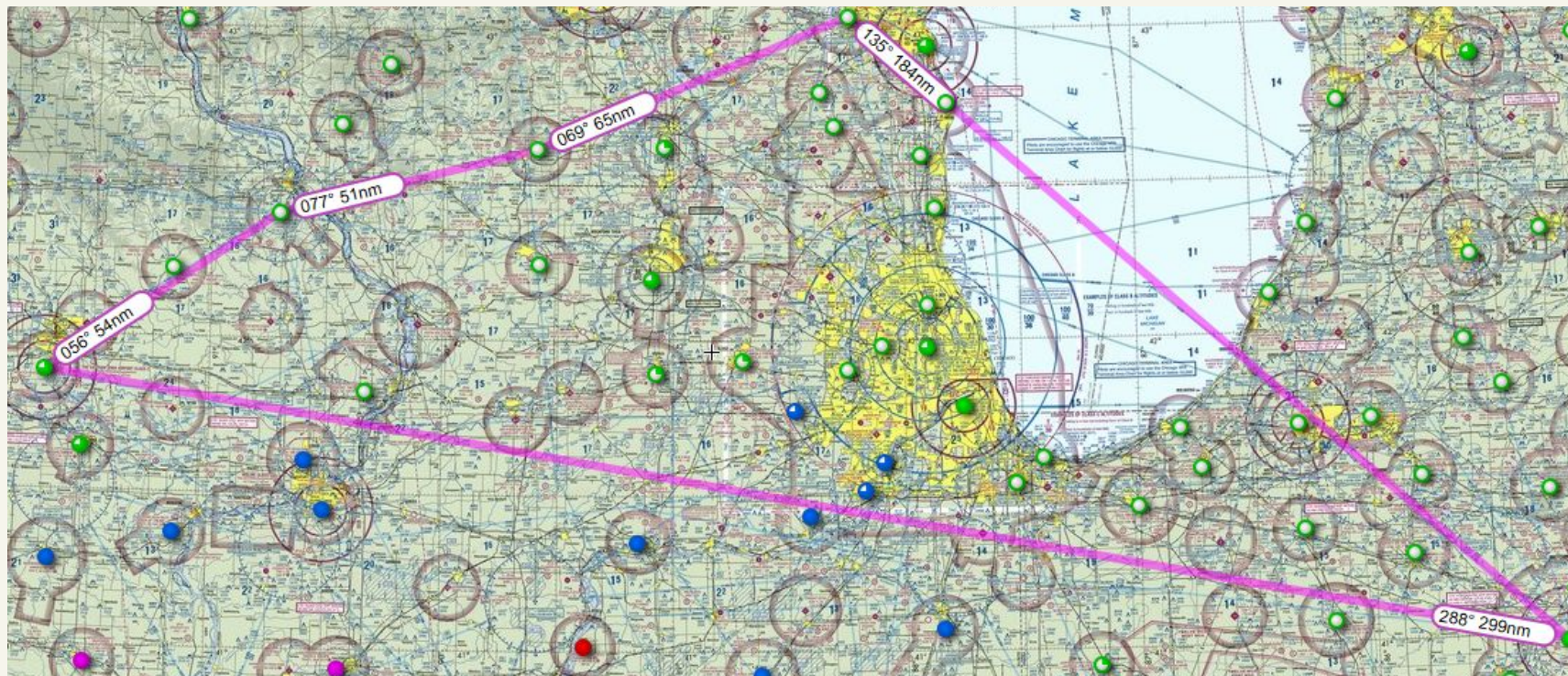
Flight Information

Departing Airport	Airport Arrival	Heading	Distance (nm)	Travel Time	Refueling needed? Yes/No
JFK	MIA	215	947.0	1.871541502	No
MIA	DEN	318	1485.0	2.934782609	Yes
DEN	MRY	254	834.0	1.648221344	No
MRY	SEA	346	652.0	1.3	No
SEA	ORD	75	1495.0	3.0	Yes
ORD	JFK	96	643.0	1.3	No



CESSNA 172 SPECIFICATIONS		
Fuel capacity	Cruising air speed	Cruising Range
44 gal	120 kt	540 nm
Airport Name	Airport Identifier	VOR Frequency
Waukesha	KUES	116.4
Fort Wayne	KFWA	117.8
Dubuque	KDBQ	115.8
Eastern Iowa	KCID	114.1

Travel Segment	Distance (nm)	Total Distance Travelled (nm)	Airspeed (kt)	Travel time (hr)	TO vector	FROM vector
Waukesha-Fort Wayne	184	184	120	1.533333333	135	315
Fort Wayne-Eastern Iowa	299	483	120	2.491666667	288	108
Eastern Iowa-Dubuque	54	537	120	0.45	56	236
Dubuque-Waukesha	116	652	120	0.966666667	72	252



CESSNA 172 SPECIFICATIONS

Fuel capacity	Cruising air speed	Cruising Range
44 gal	120 kt	540 nm
Airport Name	Airport Identifier	VOR Frequency
Waukesha	KUES	116.4
Fort Wayne	KFWA	117.8
Dubuque	KDBQ	115.8
Monroe	KEFT	114.3
Eastern Iowa	KCID	114.1

Travel Segment	Distance (nm)	Total Distance Travelled (nm)	Airspeed (kt)	Travel time (hr)	TO vector	FROM vector
Waukesha-Fort Wayne	184	184	120	1.533333333	135	315
Fort Wayne-Eastern Iowa	299	483	120	2.491666667	288	108
Eastern Iowa-Dubuque	54	537	120	0.45	56	236
Dubuque-Monroe	51	588	120	0.425	77	257
Monroe-Waukesha	65	653	120	0.541666667	69	249

Conclusions 1 and 2:

1. One symbol that was useful during flight planning was the symbol airport symbol as it allowed me to easily visualize flight paths 

Some symbols that I need to learn more about are:



2. Travel Segment	Distance (nm)	Airspeed (kt)	Travel time (hr)	Tot. Travel Time
Waukesha-Fort Wayne	184	161	1.142857143	1.142857143
Fort Wayne-Eastern Iowa	299	161	1.857142857	3
Eastern Iowa-Dubuque	54	161	0.3354037267	3.335403727
Dubuque-Monroe	51	161	0.3167701863	3.652173913
Monroe-Waukesha	65	161	0.4037267081	4.055900621

Conclusions 3, 4, 5:

3. I had to make two unexpected diversions during my flight. First, I stopped in Dubuque to refuel and arrived with just 2 nautical miles of fuel left. Then, I made another stop in Monroe due to a mid-flight emergency. These detours significantly increased my overall flight time, not even counting the ground time. Instead of a smooth triangular flight path as initially planned, the route ended up with several kinks due to these additional stops.

4. The flight plan aids the navigation system by providing all the necessary plans and calculations in advance. Having a predetermined route with all the calculations and distances already figured out helps to avoid any surprises. Essentially, the plane can use this flight plan as a guide, which in turn supports its navigation system.

5. Pilots rely on the flight plan to navigate their flights effectively. It helps them know when to tune to specific frequencies for VOR stations, where to make stops, how much fuel remains, and the planned route. The flight plan serves as a comprehensive guide, ensuring pilots know when to take specific actions for a safe and controlled flight, free from unexpected issues. Additionally, pilots use the flight plan to monitor weather conditions, helping them avoid flying into severe weather that could result in significant turbulence.